



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

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Queue Using Linked Lists

QUEUE USING LINKED LISTS

Write a C program to implement a Queue using Linked Lists for managing patients waiting in a hospital consultation room. The program should support enqueue, dequeue, display, and front operations without any fixed size limitation. Sample Input: Insert Patient IDs 201, 202, 203, Display, Delete one patient. Sample Output: Queue Elements: 201 202 203. Deleted Patient ID: 201. Remaining Queue: 202 203.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 // Structure for a Node (Patient)
5 struct Node {
6     int patientID;
7     struct Node* next;
8 };
9
10 // Structure for the Queue
11 struct Queue {
12     struct Node *front, *rear;
13 };
14
15 // Function to create a new node
16 struct Node* newNode(int k) {
17     struct Node* temp = (struct Node*)malloc(sizeof(struct Node))
18     temp->patientID = k;
19     temp->next = NULL;
20     return temp;
21 }
```

OUTPUT

Queue Elements: 201 202 203

Deleted Patient ID: 201

Remaining Queue: Queue Elements: 202 203

INPUT

Insert Patient IDs 201, 202, 203, Display, Delete one patient

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